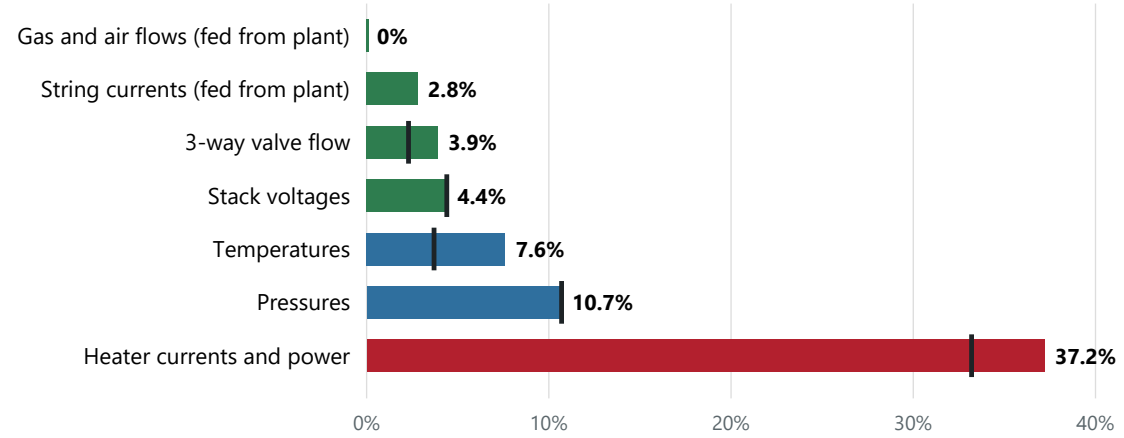
DIGITAL TWIN V32 · PART LOAD · 10 PLANT RUNS · 5 NOV 2026

In part load the twin is close on most readings, but not on all

We fed the plant's own knob settings, minute by minute, into the twin for all 10 part-load runs in the POC data and compared every one of the twin's 111 sensor readings with what the plant measured. Half of the readings miss by less than 7.7 % of that sensor's full working range, and 60 % miss by less than 10 %.

Part load in plant data	Typical miss (111 readings)	Stack temperature miss	Worst parts
268 min (10 runs, 8–44 min)	7.7 % of range	10.8 °C today, 3.9 °C with one fix	Ejector, recycle heater, reformer: 120–200 °C too cold



Typical (median) miss per group, as % of each sensor's full working range. Black tick = what a single fixed number (the plant's own part-load middle value) would score. Flows and string currents are fed in from the plant, so they match by design.

How to read this

A "miss" is the gap between the twin and the plant at the same minute, given as a share of the sensor's normal span over all stages, so a 600 °C stack and a 5-bar supply pipe can be compared.

Limits: only 268 minutes of plant part load exist, all in March 2026. The twin's part-load look-up tables were built from these same minutes, so look-up readings are checked on data they have already seen. Stack temperature and air heater are computed by physics: a fairer test.

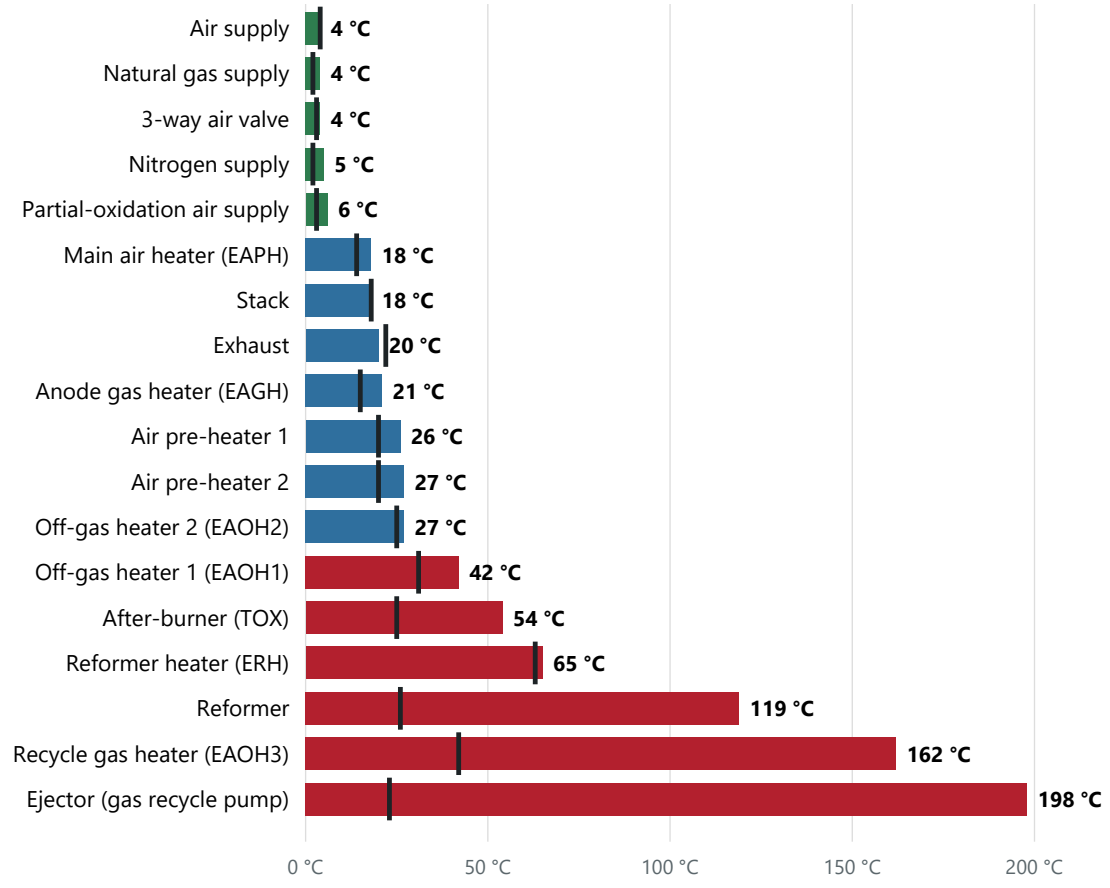
If you remember one thing: the twin's inputs and the air side are close; the gas side of the anode loop lags far behind the plant, and the stack runs about 11 °C cold because of one smoothing rule.

GROUP BY GROUP

Where the twin is close and where it is not

Group	Sensors	Typical miss	Within 10 %	In plain words
Gas and air flows (fed from plant)	6	0 %	100 %	Same as the plant: the twin is given these.
String currents (fed from plant)	5	2.8 %	80 %	The twin uses one current for all five strings. The plant's string 4 runs about 0.9 A lower (5.5 vs 6.4 A).
3-way valve flow	1	3.9 %	100 %	Within 27 m^3/h of 581 m^3/h.
Stack voltages	10	4.4 %	100 %	Within ~20–30 V of ~430–470 V. The twin shows one fixed part-load value; it does not react to current yet.
Temperatures	58	7.6 %	66 %	Typical miss 21 °C. Air side 4–27 °C; anode gas loop 120–200 °C (chart below).
Pressures	25	10.7 %	32 %	Fixed part-load values. As good as a fixed number can be: the rest is the plant's own minute-to-minute wobble.
Heater currents and power	6	37.2 %	0 %	Worst group. Heater controllers hunt on the plant (2–5 A swings); the main air heater shows 0 kW in the twin vs 8.6 kW on the plant.

Temperatures, by part of the plant



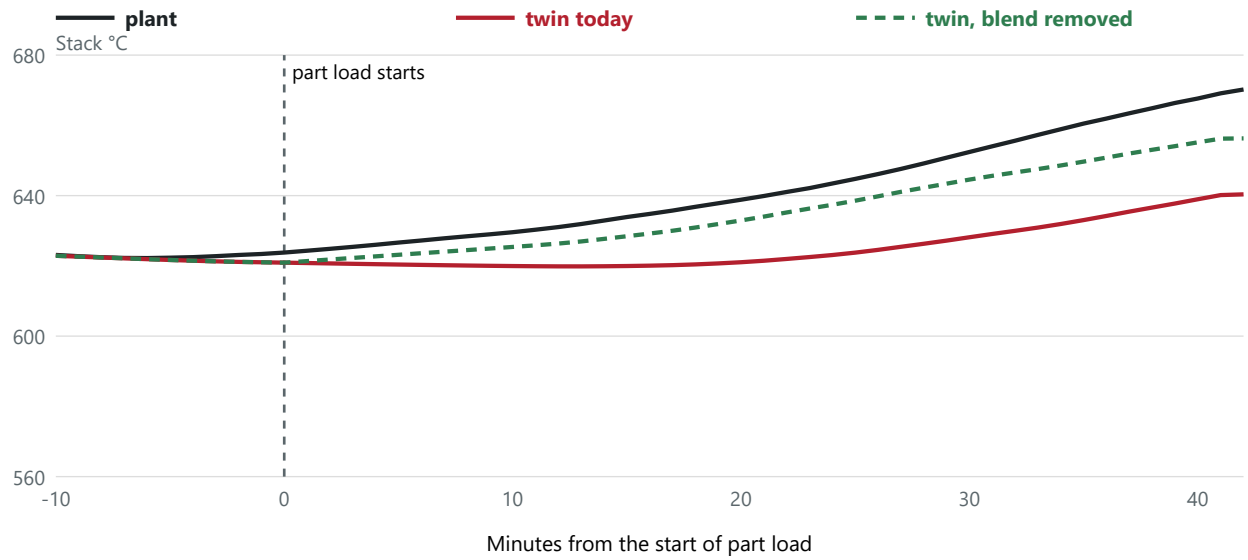
Typical temperature miss in °C per plant part (58 sensors). Black tick = what a single fixed number would miss. The three red bars are the anode gas loop. The twin reads too cold almost everywhere hot; only the reformer heater (ERH) reads too hot (+41 °C).

If you remember one thing: 57 % of the temperatures are within 25 °C of the plant, 78 % within 50 °C; the big misses sit in one place, the anode gas loop.

WHY THE MISSES

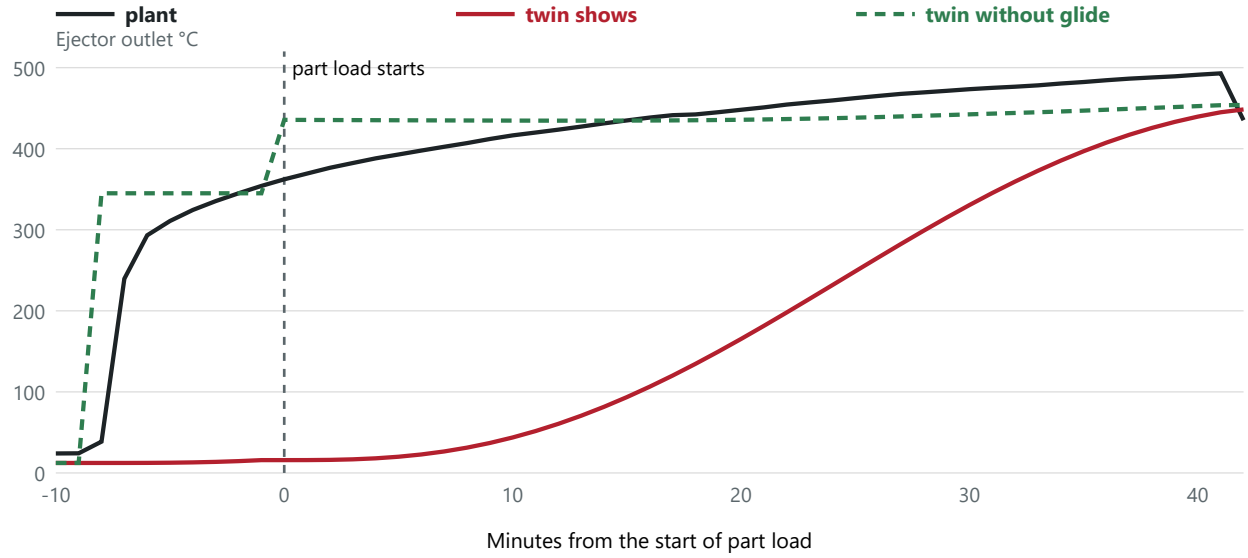
Two smoothing rules make the twin late

1 · The stack warms too late



Run of 24 March 2026, 13:54. On a stage change the twin blends its warming speed over the plant's bend time (24 min, stretched to 47 min) starting at the stage change; the plant starts bending ~10 min before the label changes. Stack miss in part load, 10 runs: today 10.8 °C, blend removed at fuel switch → part load 3.9 °C, twin v14 about 3 °C.

2 · The anode gas loop warms too late



Same run. The plant's ejector outlet jumps from ~25 to ~340 °C within ~5 min of gas on; the twin glides between stage tables over 40–60 min. Without the glide its table is within ~24 °C. Same for reformer, recycle gas heater (EAOH3) and recycle line.

3 · The main air heater shows no power

In part load the twin's recuperated air already reaches the 600 °C target, so its heater power works out to 0 kW; the plant keeps about 8.6 kW (open decision F from the earlier twin-vs-plant check).

If you remember one thing: the twin's tables and physics are mostly right for part load; the gaps come from it being too slow at the moment the plant switches to gas and load.

WHAT TO DECIDE

Five choices, in the order we would make them

	Decision	What it would change
A	Drop the warming-speed blend at fuel switch → part load (keep it elsewhere)	Stack miss 10.8 → 3.9 °C on the 10 runs. The plant's own part-load speed already contains the bend, so the blend counts it twice (same finding as the jerk test on 3 Nov).
B	Use the plant's real timing (~5 min) for gas-loop temperatures when gas comes on, instead of the 40–60 min default glide	Look-up readings: typical miss 9.3 % → about 5.6 % (76 % of them within 10 % instead of 56 %). Ejector, reformer and recycle heater from 120–200 °C off to about 20–45 °C.
C	Compute the main air heater power from the heat it puts into the air (open decision F)	Shows ~8–9 kW in part load like the plant, instead of 0 kW.
D	Give each of the 5 strings its own current	String 4 current matches (the plant runs it ~0.9 A lower).
E	Get longer part-load runs from the plant	Today there are only 268 min in 10 runs, all under 45 min, and the look-up tables were built from the same minutes. A held-back test needs new data.

Nothing has been changed in the twin yet.

If you remember one thing: decisions A and B are small changes that remove most of the gap.

ALL 111 SENSORS

Every reading, worst first

How: knob = fed from plant; model = twin physics; look-up = per-stage line vs stack temperature. % of range = miss ÷ the sensor's full working range. In band = minutes inside the plant's own middle 80 % for part load.

Tag	What it measures	How	Plant	Twin	Miss	% range	In band
IIR 3.1.1	EAGH current	look-up	4.76	4.41	2.28 A	45.7	100 %
TIRC 3.03	Ejector outlet	look-up	444	345	198 °C	40.0	13 %
TIR 3.06	Reformer inlet mixed	look-up	416	268	189 °C	39.7	19 %
IIR 6.1.1	EAOH3 current	look-up	4.21	8.45	5.11 A	38.8	63 %
IIR 5.1.1	EAOH1 current	look-up	2.97	3.36	1.96 A	37.9	100 %
IIR 5.1.2	EAOH2 current	look-up	2.90	3.13	1.91 A	36.6	100 %
TIRZ 8.2.4	EAOH3 safety (trip 950)	look-up	473	312	163 °C	30.2	14 %
TIRC 3.07	Catalyst inlet	look-up	712	613	243 °C	29.7	48 %
TIR 6.01	Recycle gas inlet	look-up	609	520	216 °C	29.6	12 %
TIR 1.3	Purge gas	look-up	13.3	9.01	5.37 °C	29.5	30 %
TIR 1.2.2	POx air FPM → PGM	look-up	17.8	15.0	6.52 °C	29.4	25 %
TIR 1.4.2	Fuel FPM → PGM	look-up	17.4	15.2	5.94 °C	28.7	44 %
TIRC 6.1.1	EAOH3 control	look-up	512	350	160 °C	28.0	16 %
EAPH power	Heater power (3-phase)	model	8.56	0.00	9.41 kW	23.6	7 %
TIR 3.04	POx air PGM inlet	look-up	19.6	13.4	5.75 °C	22.2	21 %
PIR 1.2.1	POx air supply	look-up	5.34	5.34	0.55 bar	19.1	100 %
TIR 1.2.1	POx air (FPM)	look-up	20.0	16.4	3.89 °C	16.8	76 %
PIR 1.1.3	Air stream 2 supply	look-up	4.98	4.98	0.51 bar	16.5	100 %
PIR 1.1.1	Air stream 1 supply	look-up	5.06	5.06	0.47 bar	15.7	100 %
IIR 5.1.3	ERH current	look-up	5.14	3.60	0.82 A	15.4	100 %
TIRC 5.5.1	TOX inlet anode 1	look-up	634	572	102 °C	13.8	18 %
TIR 1.1.1	Air after MFCs	look-up	21.4	19.3	3.68 °C	13.4	100 %
PIR 2.01	EAPH inlet	look-up	122	122	18.2 mbar	12.7	100 %
PIR 1.1.2	Air after MFCs	look-up	250	250	38.2 mbar	12.6	100 %
PIR 7.02	APH exhaust outlet	look-up	0.81	0.82	0.12 mbar	12.1	100 %
TIR 2.04	TWV → APH inlet	look-up	22.4	19.7	3.96 °C	12.1	100 %
PIR 5.1.1	Cathode outlet 1	look-up	19.2	19.2	2.98 mbar	11.6	100 %
PIR 5.3.1	TOX inlet cathode 1	look-up	18.1	18.1	2.81 mbar	11.6	100 %
PIR 7.03	Exhaust	look-up	0.48	0.48	0.07 mbar	11.5	100 %
PIR 2.02	Stack module air inlet	look-up	21.8	21.8	3.31 mbar	11.5	100 %
TIR 1.4.1	Fuel (FPM)	look-up	15.5	13.7	1.76 °C	11.4	87 %
PIR 1.2.2	POx air after MFC	look-up	80.9	80.9	21.4 mbar	11.1	100 %
EIRC 4.4.2	String 4 current	knob	5.54	6.41	1.45 A	11.1	66 %
TIR 2.1.2	APH2 inlet air	look-up	22.3	19.4	4.10 °C	11.0	78 %
TIR 2.1.1	APH1 inlet air	look-up	22.6	19.7	4.12 °C	10.9	77 %
TIRC 5.09	Burner (TOX) representative	look-up	741	644	87.9 °C	10.8	24 %
PIR 7.1.2	APH inlet exhaust 2	look-up	7.99	7.99	1.18 mbar	10.7	100 %
PIR 7.1.1	APH inlet exhaust 1	look-up	8.02	8.02	1.19 mbar	10.7	100 %

Tag	What it measures	How	Plant	Twin	Miss	% range	In band
PIR 1.4.2	Fuel after MFC	look-up	8.93	8.93	1.07 bar	10.7	100 %
PIR 3.01	Ejector main stream	look-up	8.84	8.84	1.07 bar	10.7	100 %
TIR 3.01	Fuel PGM inlet	look-up	16.4	18.1	3.70 °C	10.7	57 %
TIRC 5.5.2	TOX inlet anode 2	look-up	621	584	81.6 °C	10.6	28 %
PIR 5.4.1	TOX inlet anode 1	look-up	16.6	16.6	2.50 mbar	10.1	100 %
PIR 5.4.2	TOX inlet anode 2	look-up	16.7	16.7	2.50 mbar	10.1	100 %
TIR 7.3.2	Flue gas out 2	look-up	94.9	104	17.5 °C	9.8	100 %
PIR 3.03	Stack module anode inlet	look-up	22.2	22.2	2.86 mbar	9.8	100 %
PIR 5.2.1	Anode outlet 1	look-up	17.1	17.1	2.53 mbar	9.7	100 %
TIR 7.04	APH exhaust outlet	look-up	113	125	21.1 °C	9.7	100 %
PIR 6.01	Recycle gas inlet	look-up	17.0	17.0	2.49 mbar	9.5	100 %
TIRZ 8.1.9	Burner safety (trip 930)	look-up	707	628	73.2 °C	9.4	33 %
TIR 7.05	Exhaust (roof)	look-up	112	124	20.2 °C	9.3	100 %
TIRZ 8.1.10	Exhaust safety (trip 300)	look-up	114	125	19.9 °C	9.2	100 %
TIR 7.3.1	Flue gas out 1	look-up	97.8	106	16.8 °C	9.1	100 %
TIRZ 8.2.5	ERH safety (trip 860)	look-up	712	766	68.3 °C	8.2	100 %
TIR 7.1.2	TOX outlet 2	look-up	694	626	61.4 °C	8.2	37 %
TIRC 5.17	ERH control	look-up	702	749	62.5 °C	7.7	100 %
PIR 3.02	Ejector outlet	look-up	31.0	31.0	2.96 mbar	7.5	100 %
TIRC 3.15	Catalyst outlet	look-up	614	560	49.3 °C	7.4	15 %
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	look-up	664	608	49.7 °C	7.0	34 %
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	look-up	658	602	49.0 °C	6.9	38 %
TIR 7.1.1	TOX outlet 1	look-up	667	615	47.3 °C	6.4	49 %
EIRC 4.3.1	String 3 load voltage	look-up	446	446	31.5 V	6.3	100 %
PIR 1.5	H_2 supply	look-up	5.38	5.38	0.07 bar	6.0	100 %
TIRC 5.3.1	EAOH1 control	look-up	660	631	42.3 °C	5.7	78 %
TIR 3.05	Reformer inlet POx air	look-up	32.0	29.1	10.1 °C	5.7	54 %
TIR 5.6.1	Cathode bypass 1	look-up	616	575	37.5 °C	5.6	38 %
TIR 2.03	APH outlet riser → EAPH	look-up	585	549	36.1 °C	5.4	48 %
TIR 2.02	APH1+2 outlet air	look-up	585	550	36.2 °C	5.4	49 %
TIRZ 8.2.2	EAOH1 safety (trip 950)	look-up	660	633	39.6 °C	5.4	81 %
Σ stacks M3	String 3 stack-sum voltage	look-up	422	422	25.1 V	5.1	100 %
EIRC 4.2.1	String 2 load voltage	look-up	460	460	25.6 V	5.1	100 %
EIRC 4.1.1	String 1 load voltage	look-up	455	455	24.6 V	4.9	100 %
TIRC 3.17	Reformer outlet	look-up	619	583	32.1 °C	4.8	36 %
EIRC 4.5.1	String 5 load voltage	look-up	470	470	24.7 V	4.8	100 %
TIR 3.02	Ejector main stream	look-up	22.5	23.6	3.06 °C	4.7	59 %
TIRC 5.2.1	Anode outlet 1	look-up	648	632	28.2 °C	4.0	86 %
Σ stacks M2	String 2 stack-sum voltage	look-up	434	434	19.5 V	3.9	100 %
FIRC 1.1.3	3-way valve (temperature-driven)	look-up	581	551	23.9 m^3/h	3.9	56 %
TIR 2.05	EAPH inlet	model	582	600	24.9 °C	3.9	39 %

Tag	What it measures	How	Plant	Twin	Miss	% range	In band
Σ stacks M1	String 1 stack-sum voltage	look-up	430	430	18.8 V	3.9	100 %
PIR 1.3.1	N_2 supply	look-up	5.52	5.52	0.05 bar	3.7	100 %
EIRC 4.4.1	String 4 load voltage	look-up	454	454	18.4 V	3.7	100 %
Σ stacks M5	String 5 stack-sum voltage	look-up	444	444	17.9 V	3.5	100 %
TIR 3.19.1	Manifold 1 inlet	look-up	601	582	21.2 °C	3.3	50 %
TIRC 5.3.2	EAOH2 control	look-up	658	646	26.6 °C	3.3	100 %
TIR 3.18	Stack module anode inlet	look-up	616	596	22.1 °C	3.3	54 %
TIRZ 8.2.1	EAGH safety (trip 950)	look-up	613	597	21.4 °C	3.2	63 %
TIRC 3.1.1	EAGH control	look-up	610	596	20.6 °C	3.2	66 %
TIR 3.19.2	Manifold 2 inlet	look-up	597	579	19.7 °C	3.1	51 %
TIRZ 8.2.3	EAOH2 safety (trip 950)	look-up	657	648	23.9 °C	3.0	100 %
EIRC 4.5.2	String 5 current	knob	6.31	6.41	0.47 A	2.9	84 %
EIRC 4.3.2	String 3 current	knob	6.47	6.41	0.46 A	2.8	85 %
TIRZ 8.1.3	Stack safety (trip 720)	look-up	642	620	19.1 °C	2.8	72 %
Σ stacks M4	String 4 stack-sum voltage	look-up	428	428	13.3 V	2.8	100 %
EIRC 4.2.2	String 2 current	knob	6.34	6.41	0.44 A	2.6	85 %
TIRZ 8.1.1	EAPH safety (trip 1050)	look-up	592	591	17.6 °C	2.6	100 %
TIRC 5.2.2	Anode outlet 2	look-up	643	623	17.6 °C	2.6	100 %
TIR 5.4.1	TOX inlet cathode 1	look-up	622	606	14.6 °C	2.2	78 %
TIRC 5.1.2	Cathode outlet 2	look-up	628	613	14.2 °C	2.1	83 %
TIRC 5.1.1	Cathode outlet 1	look-up	624	609	13.5 °C	2.0	89 %
TIR 4.6.x	Stack median of 36 TCs (model)	model	625	612	12.1 °C	1.8	82 %
EIRC 4.1.2	String 1 current	knob	6.51	6.41	0.28 A	1.7	85 %
TIRC 2.06	EAPH outlet = stack air in	model	600	600	5.71 °C	0.9	86 %
PIR 1.3.2	Purge gas	look-up	1.83	1.83	0.11 bar?	0.5	100 %
PIR 1.4.1	After booster	look-up	13.4	13.4	0.04 bar	0.3	100 %
FIRC 1.1.1	Cathode air stream 1	knob	4500	4500	2.15 ln/min	0.0	77 %
FIRC 1.1.2	Cathode air stream 2	knob	4500	4500	2.15 ln/min	0.0	79 %
FIRC 1.2	POx air	knob	312	312	0.00 ln/min	0.0	80 %
FIRC 1.5	Hydrogen	knob	0.00	0.00	0.00 ln/min	0.0	99 %
FIRC 1.3	Nitrogen	knob	0.00	0.00	0.00 ln/min	0.0	99 %
FIRC 1.4	Natural gas	knob	137	137	0.00 ln/min	0.0	79 %

If you remember one thing: 36 % of readings are within 5 % of range and 86 % within 20 %; the worst ten are heater currents and anode-loop temperatures.